

- 11-4 Find the driving-point impedance seen by the voltage source in Figure P11-4 and the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$.

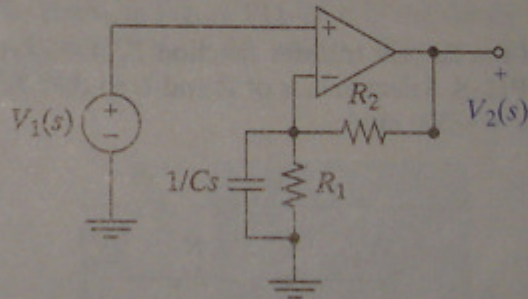


FIGURE P11-4

- 11-7 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ in Figure P11-7. Select values of R and C so that $T_V(s)$ has a pole at $s = -500$ rad/s.

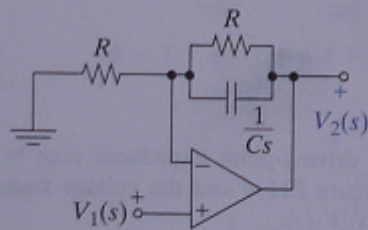


FIGURE P11-7

- 11-10 Find the voltage transfer function $T_V(s) = V_2(s)/V_1(s)$ of the cascade connection in Figure P11-10. Locate the poles and zeros of the transfer function.

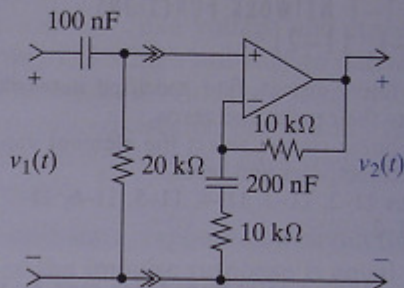


FIGURE P11-10

- 11-29 The circuit in Figure P11-29 is in the steady state with $i_1(t) = 5 \cos 1000t$ mA, $R = 1 \text{ k}\Omega$, $L = 2 \text{ H}$, and $C = 500 \text{ nF}$. Find $i_{2ss}(t)$. Repeat with $i_1(t) = 10 \cos 2000t$ mA.

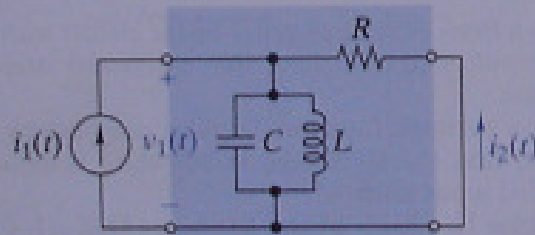


FIGURE P11-29

- 11-53 Design a circuit to realize the transfer function below using only resistors, capacitors, and not more than

one OP AMP. Scale the circuit so that the final design uses only 20-k Ω resistors.

$$T_V(s) = \pm \frac{1000 s}{(s + 500)(s + 1000)}$$

- 11-54 Design a circuit to realize the transfer function below using only resistors, capacitors, and inductors (no OP AMPs allowed). Scale the circuit so that all inductors are 50 mH or less.

$$T_V(s) = \frac{s^2}{(s + 2000)(s + 4000)}$$

- 12-2 Find the transfer function $T_V(s) = V_2(s)/V_1(s)$ of the circuit in Figure P12-2.

(a) Find the dc gain, infinite frequency gain, and cutoff frequency. Identify the type of gain response.

- (b) Sketch the straight-line approximations of the gain response.
- (c) Calculate the gain at $\omega = 0.5\omega_C$, ω_C , and $2\omega_C$.

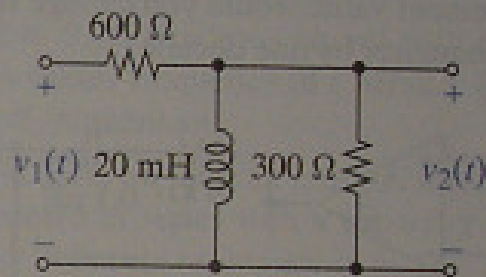


FIGURE P12-2

12-10 Repeat Problem 12-9 for

$$T(s) = \frac{10/s}{100/s + 1/20}$$

12-14 The circuit in Figure P12-14 produces a bandpass response for a suitable choice of element values. Identify the elements that control the two cutoff frequencies. Select the element values so that the passband gain is 10 and the cutoff frequencies are 100 rad/s and 2500 rad/s. Use practical element values with $R \geq 10 \text{ k}\Omega$ and $C \leq 1 \mu\text{F}$.

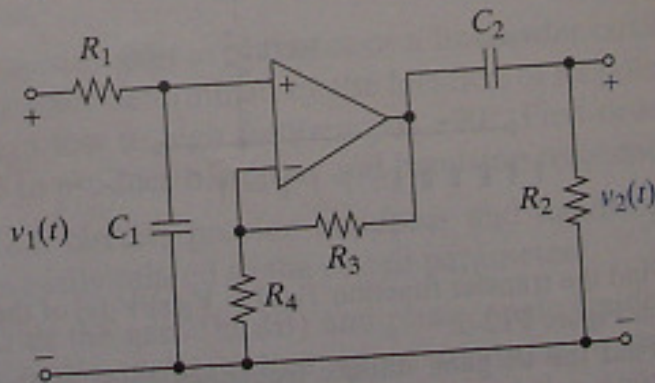


FIGURE P12-14

12-54 **E** Chip RC Networks

Integrated circuit (chip) RC networks are used at parallel data ports to suppress radio frequency (RF) noise. In a certain application RF noise at 3.2 MHz is interfering with a four-bit parallel data signal operating at 1.1 MHz. A chip RC network is to be used to reduce the RF noise on the parallel bus by at least 7 dB without reducing the data signals by more than 2 dB. A vendor offers a family of chip RC networks connected as shown in Figure P12-54. The available circuit parameters are shown in Table P12-54. Select the part number that best meets the noise suppression requirements.

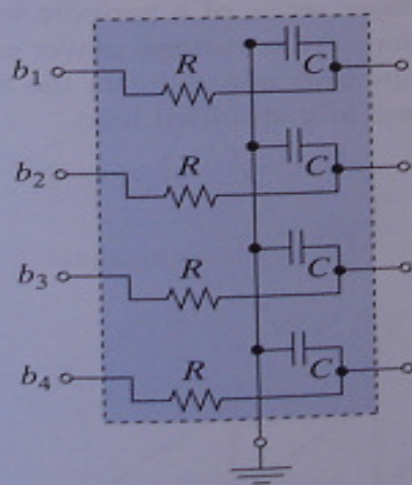


FIGURE P12-54